CMSE 202 Project Final Report

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What Makes a Song Popular on Spotify?

In this project, we aimed to answer a simple yet interesting question: what makes a song popular on Spotify? With millions of users and a huge catalog of music, Spotify tracks many audio features and assigns each song a popularity score between 0 and 100. We wanted to know which features—danceability, energy, tempo, or loudness, for example—go into making a song more popular. To do this, we used a Kaggle dataset that included over 160,000 tracks from the year 2006 to 2020, with each track including extensive information about various music-related aspects. We first imported required Python libraries like Pandas, Seaborn, Matplotlib, and Scikit-learn, then filtered the data to include songs from the year 2006 and beyond. We inspected the data structure and cleaned the data by removing any missing values so that we could ensure our analysis would be valid. With the cleaned data, we used histograms to understand the distribution of popularity scores. We observed that most songs had an average popularity score in the 60s, with very few songs having very low or very high scores. We also created a correlation heatmap to observe how popular each feature was related to popularity. We observed that more recently released songs were more popular, and features like energy and loudness were also positively correlated moderately. Acousticness had a reverse relation with popularity, i.e., more acoustic songs were less popular on average. We used scatter plots to examine how each feature alone—like danceability, valence, and speechiness—might impact popularity, and while some showed more distinct patterns than others, we noticed that more energetic and happier songs were more likely to fall on the higher side of the scores. To play around with how well we could predict a song's popularity, we built a simple linear regression model. We selected features like energy, loudness, danceability, valence, and tempo as inputs, and popularity as the output. We then divided the dataset into a training and a test set, trained the model, and R^2 score and Mean Squared Error were used to evaluate the performance of the model. While the model didn't predict with absolute accuracy, it did pick up on a good portion of the trend and gave us some fascinating insight into which song features are helpful in determining popularity. Overall, we discovered that while external variables like artist popularity or social media following also play a role in the success of a song, musical features—especially energy, loudness, and the newness of a song—do have a measurable effect. This project opened our eyes to the power of using data to identify trends in the music industry, and in the future, we could expand the model to include other features like genre, artist popularity, or playlist position to allow predictions to be even more accurate.